

AI in Companies' Production Processes: What Do Their CEOs Think?

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ABSTRACT

The accelerated integration of Artificial Intelligence (AI) in comprehensive organizational management has marked a significant milestone in enhancing efficiency and productivity across all sectors. However, the effective adoption of this emerging technology faces significant challenges, such as ethical dilemmas, organizational barriers, and a notable deficit in relevant technological skills. This study embarks on a detailed analysis of the crucial determinants influencing the adoption of AI by companies, enhancing the UTAUT model with four new variables: Response Costs, Trust in AI, AI Anxiety, and Environmental Sustainability. Through surveys directed at over 400 CEOs of companies, this work reveals that facilitating conditions, performance expectancy, response costs, trust in AI, and AI anxiety determine the adoption of this new technology in their companies. These findings contribute to identifying which factors, from a managerial perspective, should be considered as more than sufficient reasons for AI to be implemented in their production processes.

KEYWORDS

Artificial Intelligence (AI), Internal Processes, Production Processes, AI Implementation, Response Costs, AI Anxiety, Trust in AI, Environmental Sustainability

AI IN COMPANIES' PRODUCTION PROCESSES: WHAT DO THEIR CEOS THINK?

The adoption AI in companies' internal and productive processes has significantly impacted multiple sectors worldwide (Gonzalez et al., 2023; Huang & Rust, 2018). From energy efficiency in manufacturing (Liu et al., 2022) and organizational innovation (Mariani et al., 2023) to project management and logistics in construction (Na et al., 2023), AI has proven to be a critical component for process improvement and outcome optimization (Sangeetha et al., 2022). AI can be used for various purposes, such as forecasting, optimization, and automation (Srivastava et al., 2024). Additionally, AI methods like machine learning algorithms have been used for customer modeling, which has contributed to the personalization and enhancement of consumer experiences (Gupta & Panigrahi,

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2023). A recent study by McKinsey & Company (2022) highlights that AI has emerged as a vital technology that companies must incorporate to plan their future and improve productivity.

Although AI has shown robust potential for process automation (Aysolmaz et al., 2023), inventory optimization (Maldonado-Canca et al., 2024; Mekhanikov, 2022) and improving resource management efficiency in marketing (Reshak & Abdulridha, 2023), some authors, like Parteka and Kordalska (2023), question whether its application directly translates into increased business productivity. The implementation of AI is not without obstacles, such as ethical dilemmas, organizational barriers, and technological and technical challenges (Vakkuri et al., 2022; Zirar et al., 2023), as well as concerns about algorithmic biases and the handling of confidential data (Agrawal et al., 2022; Dwivedi et al., 2021). Furthermore, from the perspective of human resources policies for this technological adoption, while the use of AI in companies promotes the development of effective professional skills (Mikalef et al., 2023), a lack of technological understanding and a resistance to change emerge as challenges requiring a solid and detailed approach (Khan et al., 2023).

In this context, knowing the attitude of the main decision-maker in an organization—the CEO—is key to ensuring not only the implementation of this technology but also its continuity. According to Wang and Wu (2024), the CEO's attention to innovation influences organizational focus, resource allocation, and decision-making processes on technology adoption. According to several authors (Li et al., 2021; Pinski et al., 2024), CEOs' orientations towards AI have introduced a new impetus to organizational decision-making and strategic direction. Specifically, CEO emphasis on AI fosters process innovation within firms (Füller et al., 2022; Rammer et al., 2022), decreasing resource consumption and offering opportunities for organizational efficiency in innovation efforts (Maliha et al., 2021), with one of the most revealing currently focused on green innovation (Liu & Zhou, 2024; Wang et al., 2024). The use of AI in resource optimization and supply chain management enables the redeployment of resources towards green innovation initiatives, such as green product development and sustainable manufacturing practices (Almansour, 2023; Benzidia et al., 2021; Ying & Jin, 2024). On the other hand, even if CEOs perceive long-term innovation as valuable in their organizations, they are nevertheless aware that they must balance short-term cost reduction to increase competitive advantage and the survival of their firms (Lok et al., 2024). In this context, response costs—understood as cost of effort in terms of time, workload, and money—may motivate CEOs to adopt AI. Indeed, the work of Wang et al. (2022) shows that managers of companies in China are more sensitive to what they spend on adopting new technology than those in developed countries. Moreover, the time spent on adoption is associated with opportunity cost. The authors also revealed that there is a psychological cost for company managers linked to concerns about losing money. This mental suffering is related to anxiety or fear of failure in regards to this implementation. Finally, just as numerous studies contemplate the importance of trust for any user in the intention to adopt new technologies (McLean et al., 2020; Pillai & Sivathanu, 2020a; Zhu et al., 2023), including AI (Choung et al., 2023), in the case of the intention of CEOs towards AI's technological adoption, it may be marked by their confidence towards it (Khalil & Zainuddin, 2015); thus for example, among others it becomes a determinant in the intentions of public sector managers (Majrashi, 2024).

Although the literature explaining the drivers and barriers to the adoption of new technologies in companies has been widely studied, with a large number of theories and adoption models, there are two major gaps: studies on the implementation of technologies such as AI offer a general perspective on technological adoption in companies without delving into the vision of those who make the decision and ensure its implementation and continuity; and the factors studied so far are standard when explaining any technological adoption without considering other more specific factors derived from the concerns that CEOs might have when adopting AI in the productive processes of their organization. Given this debate, our aim is to provide a more comprehensive and insightful perspective on the challenges and opportunities that AI implementation presents for companies in their production processes. We seek to understand the perceptions of the CEOs. Specifically, the research questions guiding this study is:

- RQ1:** What are the determining factors in the adoption of AI tools in internal processes from the perspective of CEOs?
- RQ2:** What are the specific drivers and barriers influencing CEOs' decisions to adopt AI in production environments?

This research approach will offer insights to those implementing this technology in companies concerning which obstacles or drivers identified by CEOs could slow down, accelerate, or ensure the sustainable and reliable growth of AI in the processes of organizations. To this end, this paper presents the theoretical justification and relationships of the proposed model, its methodology, the results, the discussions, and the conclusions, highlighting the main contributions, implications, and limitations of the research.

THEORETICAL JUSTIFICATION AND RELATIONSHIPS OF THE PROPOSED MODEL

The accelerated introduction of AI into companies' productive and internal processes underscores the imperative need for a robust theoretical framework to analyze and predict the effective adoption of this transformative technology (Hewage, 2023). In this context, the traditional technology acceptance model (TAM), although widely used, proves insufficient (Ullah et al., 2023). The TAM has shown significant limitations in predicting the adoption of complex technologies, demonstrating a reach that does not exceed 50% of cases (Park, 2009).

The multidimensional nature of AI requires a model that integrates key acceptance theories, encompassing factors influencing its adoption. The unified theory of acceptance and use of technology (UTAUT) model appears to be suitable for two reasons. First, it overcomes previous limitations by incorporating elements from eight influential theories and models: the theory of reasoned action, the diffusion of innovation theory, the theory of planned behavior, the TAM, the combined TAM and theory of planned behavior model, the motivational model, the model of PC utilization, and the social cognitive theory (SCT). Second, recent studies (Alkhwalidi & Abdulmuhsin, 2022; Horodyski, 2023) reaffirm the UTAUT model's ability to identify significant predictors in the acceptance of advanced technologies, such as AI, by companies (Fu et al., 2024). In this sense, this model not only considers performance expectancy and effort expectancy, but also considers social influence and facilitating conditions as key determinants (Venkatesh et al., 2003) of adoption.

The UTAUT model has proven to be adaptable and useful in organizational contexts (Kwarteng et al., 2023). Recent studies have expanded its application to include relevant factors at the business level, such as competitive pressure and facilitating conditions, which are crucial in the adoption of digital technologies by small and medium-sized enterprises (SMEs) (Alhaimer, 2019; Henao-Ramírez & López-Zapata, 2022). For example, Alrousan et al. (2021) found that market pressure drives SMEs to adopt digital technologies; they found that facilitating conditions are key in the decision to adopt technologies such as social media for electronic advertising (Alhaimer, 2019).

In line with what was mentioned in the introduction, contexts related to CEO concerns require the inclusion of additional specific factors to the four existing ones in the UTAUT model to better address the idiosyncrasies and challenges of implementing AI in companies' internal management processes. As stated by Al-Sharafi et al. (2023), although the UTAUT model has been widely used to assess users' behavioral intentions regarding the use of emerging innovations, it does not take into account how users' fears and coping assessments may affect their behavioral intentions. In the context of AI, its ability to predict CEOs' intentions may be limited. In order to improve this ability, the UTAUT model is taken as a basis and enriched with new factors extracted from other proven theories. Among the most effective explanatory theories is the perception motivation theory, developed by Rogers (1975), which proposes that threat and coping appraisals are the main determinants of individuals' protective behaviors (Rogers, 1983). The perception motivation theory is widely recognized for its ability to

predict an individual's intention to adopt preventive measures in response to technological change (Zhang et al., 2020). In our model, the concept of coping evaluations is operationalized through the construct of response costs, which reflects the perceived effort, time, and financial burden associated with adopting AI technologies.

On the other hand, CEOs' reactions to new technology may be derived, as for any employee, from their emotional experiences with it (Stam & Stanton, 2010). Affect is central to individuals' intentions and attitudes to adopting any technology (Venkatesh et al., 2003). However, when what occurs is a negative emotion, anxiety and a withdrawal from the intention to use manifest (Chan et al., 2024). In line with this, the model is reinforced with the SCT in order to know if the adoption and use of AI depends on the CEOs being convinced that they have the skills to succeed in its implementation or use. That is, SCT puts people's beliefs about their ability to perform a specific task and achieve success at the center. Psychological states, such as anxiety, affect the confidence we have in our abilities, reducing our self-efficacy. However, controlling potential threats plays a central role in the activation of anxiety (Bandura, 2005). Bandura (2005) states that threat in something, as it may be in the technological adoption of AI, is a relational property that reflects the correspondence between perceived coping capabilities, as may be in the case of the present study the CEO's perceived response costs. Based on the approaches of the planetary boundaries (PBs) (Steffen et al., 2015) and the sustainable development goals (SDGs) (United Nations, 2015), the model is enriched with the environmental sustainability construct. In the case of the PBs they focus on earth system processes—in other words, they are concerned with impacts on the natural environment and ways to assess them. For the SDGs in addition to their strong focus on achieving environmental sustainability, emphasis is placed on strengthening technology transfer, building in developing countries, and promoting local public awareness to facilitate this (Dong & Hauschild, 2017).

Finally, trust is another factor that plays a key role in the adoption of very different AI technology applications (Söllner et al., 2016). Previous research on service chatbot applications inspired by human-robot interaction literature identifies trust as an imperative predictor of willingness to accept their use (Seo & Lee, 2021; Weitz et al., 2019). In this model this construct is extrapolated to any other AI application related not only to chatbot services but also to other organizational processes where this technology can be applied (Majrashi, 2024), because as several authors have already warned, the trust of the individual is key to generating changes in attitude, experience, and behavioral outcomes (Chi et al., 2021; Tussyadiah et al., 2020).

Therefore, in line with the above, four new factors in the AI adoption model for CEOs are incorporated: AI anxiety (Gaudiello et al., 2016; Suseno et al., 2022), environmental sustainability (Baudier et al., 2020; Gansser & Reich, 2021), response costs (Al-Emran et al., 2020; Johnston & Warkentin, 2010), and trust in AI (Kreiter, 2023; McKnight et al., 2011). The proposed model, therefore, comprises a total of eight variables—four factors from UTAUT models and the four new factors of the aforementioned theories—which are developed below.

Factors From UTAUT Models

Behavioral Intention

According to Venkatesh et al. (2003), behavioral intention is an individual's planned predisposition towards using a technology, a central concept in our analysis of CEOs' adoption of AI tools in business environments. Jameel et al.'s (2023) research highlights the relevance of this intention as a reflection of behavioral decisions in the incorporation of AI-based chatbots. Keszey (2020) emphasizes the importance of behavioral intention in studying behavior, providing guidance on its application in research. Chatterjee et al. (2020) reinforce the idea of focusing on usage intention, highlighting how prior perceptions and attitudes influence the successful adoption of AI tools and underscoring the need to understand initial attitudes before the practical use of this technology.

Performance Expectancy

Performance expectancy is measured through CEOs' perceptions of how AI can improve their companies' efficiency, production, and management. This variable is considered a critical element in the adoption of emerging technologies like AI in companies' internal production processes (Jain et al., 2022). Companies that anticipate significant improvements in efficiency, production, and management through the implementation of AI are more inclined to adopt this technology (Hanapiyah et al., 2022; Sharma et al., 2022). The perception that AI can enhance and optimize internal processes acts as a significant driver of its adoption (Alonso, 2023; Ferri et al., 2023). Based on robust evidence linking performance expectancy to the adoption of technological innovations, we formulate the following hypothesis:

H1: Performance expectancy is positively related to CEOs' intention to adopt AI tools in their companies.

Effort Expectancy

The adoption of AI, like other technologies, is also influenced by effort expectancy (Alshamsi, 2023; Atwal et al., 2021). If companies anticipate that the implementation of AI will be a smooth and uncomplicated process, they will be more willing to integrate this technology into their internal operations (Wang & Su, 2021). Conversely, the perception that AI tools are complex and difficult to manage can act as a significant barrier to their adoption (Bauer et al., 2020). Based on previous evidence on this factor, we propose the following hypothesis:

H2: Effort expectancy is negatively related to CEOs' intention to adopt AI tools in their companies.

Social Influence

Social influence also plays an important role in the decision to adopt new technology like AI (Bodea et al., 2020; Ferri et al., 2023). This variable measures the perceived pressure from other important actors, such as competitors and leading companies, that have already implemented AI technologies. The perception that other social agents, competitors, or leading companies are successfully implementing AI technologies can positively influence a company's willingness to adopt AI in its own internal processes (Al-Emran et al., 2020; Bodea et al., 2020; Ikumoro & Jawad, 2019). Therefore, we propose the following hypothesis:

H3: Social influence is positively related to CEOs' intention to adopt AI tools in their companies.

Facilitating Conditions

Facilitating conditions, such as the availability of technological resources and technical support, are essential for the successful implementation of AI in companies (Al-Emran et al., 2020; Dutta Gupta & Guha Majumder, 2023). Companies that perceive they have the necessary support and resources for AI implementation are more inclined to adopt this technology (Behl et al., 2022; Cao et al., 2021; Sood et al., 2022). Consequently, we propose the following hypothesis:

H4: Facilitating conditions are positively related to CEOs' intention to adopt AI tools in their companies.

New Factors in the AI Adoption Model for CEOs

Response Costs

Response costs include the efforts, time, and difficulties associated with engaging in a particular behavior (Floyd et al., 2000)—in our case, AI implementation (J. E. Chen et al., 2023; Talaei et al., 2024). These costs exert a negative impact on users' mindsets and evaluations (Rogers & Prentice-Dunn, 1997; Zhang et al., 2020). In an organizational context, when the response costs associated with technologies are perceived as being high, they become a significant barrier to adoption (Al-Emran et al., 2020; L. Chen et al., 2023; Johnston & Warkentin, 2010; Tsang et al., 2024; Wu, 2020). The same could be true for AI. However, according to Raddatz et al. (2020), the results of works analyzing the impact of response cost on an individual's intention to adopt information technology-related behaviors are inconsistent and require further research. Thus, the inhibitory or non-inhibitory role of response cost on behavior may vary depending on the context (Hanus & Wu, 2016; Menard et al., 2017). Thus, in one workplace employees' adoption of technology may not be affected by consideration of the number of resources they are expected to spend on this behavior (Ifinedo, 2012). In contrast, in other more novel contexts and with higher functional complexity of technologies, such as blockchain-based applications, understanding what is needed to preserve personal data may require much mental effort and time, making this response cost negative for the adoption intention (Marikyan et al., 2022). Similarly, in other applications, such as contactless payments using mobile apps, the response cost additionally has a negative impact on the intention to continue using these payment methods (Hamzah, 2024).

It seems logical that the acquisition and operation of AI entail inherent costs, which could deter the adoption in organizations due to increased expenses and necessary effort (Milne et al., 2000; Rainear & Christensen, 2017). However, this statement should be considered quarantined because although widely used in technology adoption, there are still insufficient studies that have addressed the impact that this variable has on the intention to adopt AI. Only a few studies, such as Fu et al. (2024) and Hung and Chen (2023), framed in a non-business environment—specifically in the academic-university environment—show the negative impact of response cost on the intention to use an AI chatbot, ChatGPT, for academic tasks.

In view of the above, it would be interesting to know whether or not the response cost affects the intention to adopt this technology in organizations and whether it is a clear barrier in CEOs' perceptions. Therefore, we propose the following hypothesis:

H5: Response costs are negatively related to CEOs' intention to adopt AI tools in their companies.

Trust in AI

When technology is considered easy to use, risk-free, and credible, in the case of the new technology, both users (Mcknight et al., 2011) and top managers (Majrashi, 2024) may come to rely on it for adoption. There are numerous works where the importance of perceived trust in very diverse organizational processes and services is confirmed. In this sense, its positive influence is considered, for example, in mobile payment services (Liébana Cabanillas et al., 2020) or, as in the case of our study, in the acceptance of the use for productive processes (Mesbah et al., 2019). In parallel, several studies analyze the influence of the trust factor on the intention to use AI tools (Choung et al., 2022; Kreiter, 2023; Lee et al., 2018). Specifically, the literature contemplates trust in AI for very different organizational processes and services, from work on trust in advisory systems (Mesbah et al., 2019), recommendation agents (Choi et al., 2011), and conversational agents (Laumer et al., 2019). One of the areas that has been most addressed in the AI literature is robotic service processes, given that replacing humans with chatbots can generate changes in user attitude, experience, and behavioral outcomes (Chi et al., 2021; Tussyadiah et al., 2020). In the organizational context, employees' belief

that such a tool works efficiently evokes a feeling of trust during the process development (Hasan et al., 2022; Wirtz et al., 2018). In addition, Mesbah et al. (2019) claim that users gain confidence mainly when they are given the opportunity to test technology in a protected environment and learn how a device works. For other authors, the social presence of this technology increases trust and reuse intentions (Choi et al., 2011) or even leads to a higher perception of usefulness and quality of service (Benbasat & Wang, 2005). Although numerous studies address contextual factors around trust in AI (Almuraqab et al., 2024; Alsheibani et al., 2019; J. Chen & Wang, 2024; Majrashi, 2022; McKnight et al., 2002; Shin, 2020), including the use of AI tools in internal processes—as stated by Bedué and Fritzsche (2022)—approaches on how to effectively manage AI in organizational practice are scarce.

Although the work of the latter authors considered that the adoption of AI could be determined by increasing trust in AI through access to knowledge, transparency, explicability, fairness, accountability, auditability, certification, privacy, security, as well as self-imposed standards and guidelines, to our knowledge there is almost no work that addresses such relevant aspects as the trust that an organization's decision makers place in adoption intention. Only the recent work by Majrashi (2024) shows that the perceived confidence of public sector managers in this technology influences its adoption in the workplace. Taking this into account, we propose the following hypothesis:

H6: Trust in AI is positively related to CEOs' intention to adopt AI tools in their companies.

AI Anxiety

Anxiety is defined as the degree of apprehension or tension an individual feels when using a particular technology (Bandura, 2005) that may be too complex to manage and whose benefits are overshadowed by its required effort (Powell, 2013). Technology anxiety is currently defined in terms of barriers for organized processes within the various subsystems of an organization. Technology anxiety is not just a theoretical concept; it fits with the actual experience of companies in the era of the fourth industrial revolution (Sajdak & Młody, 2024). Anxiety is an inherent part of functioning in complex environments such as enterprises (industrial and service) and has been extensively analyzed in widely varying adoption technologies, such as the metaverse (Hasani et al., 2024; Pillai et al., 2024), chatbots (Pillai & Sivathanu, 2020a), ChatGPT (Budhathoki et al., 2024), social robots (Modliński et al., 2024), and AI (Suseno et al., 2022)—as well as in very diverse sectors such as telemedicine (Jeng et al., 2022), mobile-taxi (Ooi et al., 2020), and agriculture (Pillai & Sivathanu, 2020b). In some circumstances, it may be a motivator for action; however, several studies suggest that the anxiety generated by an emerging and uncertain technology like AI could limit intentions to adopt it (Budhathoki et al., 2024; Gaudiello et al., 2016; Thatcher & Perrewé, 2002). A situation in which technological anxiety becomes a permanent organizational pathology is undesirable and effectively prevents the development of the company and makes it difficult to maintain a permanent competitive advantage (Sajdak, & Młody, 2024). These authors, mentioned above, state that this technological anxiety in organizations is caused by aspects such as staff competence, fear of failure, resistance to change, lack of qualified personnel, information technology infrastructure, cybersecurity, management commitment and support, compatibility and integration in the value chain, and organizational and process changes.

In the business context, it is likely that the perception of a loss of control and autonomy in decision-making due to AI causes anxiety and discomfort among decision-makers (Suseno et al., 2022). Being a relatively new and disruptive technology, AI could generate anxiety related to its unforeseen impact on business processes (Budhwar et al., 2022). Although anxiety has been used to measure worker well-being, its influence on organizational performance and job satisfaction has been the subject of scrutiny among researchers (Peccei & Van De Voorde, 2019). However, there is a paucity of research on people's perspectives on their anxiety about AI within their companies. In this regard, as stated by Hasani et al. (2024), the exploration of vital emotional factors such as managers'

anxiety towards AI in the context of technology adoption in organizations remains limited in existing studies. Scarce works such as Suseno et al. (2022) address this perspective. Specifically, this one addresses the negative impact that managers' anxiety has on the readiness to change.

Much of the current discussion by top managers revolves around whether or not to place confidence in AI technologies for processes in sectors such as medicine, and many of them experience a sense of helplessness and anxiety as they lack clarity on how to manage the emergence of AI (Hengstler et al., 2016). In this context, it seems relevant to know what CEOs feel when using AI tools because higher levels of perceived anxiety towards AI implementation could inhibit the acceptance of this technology. Therefore, we propose the following hypothesis:

H7: AI anxiety is positively related to CEOs' intention to adopt AI tools in their companies.

Environmental Sustainability

Environmental sustainability evaluates how the adoption of AI can contribute to more sustainable practices within the company (Huang et al., 2024; Nishant et al., 2020). Its link to AI is crucial, as incorporating sustainable practices can significantly optimize low-impact transportation, waste management, and other essential aspects of environmental sustainability (Habila et al., 2023). According to various authors (Dhar, 2020; Schwartz et al., 2020), green AI promotes emission reductions without increasing computational costs. It is an essential actor in guiding decision-makers towards environmentally friendly practices, proving invaluable for ensuring a sustainable future (Habila et al., 2023; Schwartz et al., 2020). However, despite the clear potential of green AI to move organizations towards more eco-friendly practices, research on its impact on usage intention and adoption, under external pressures, remains scarce (Fecho & Wahl, 2023).

The important role of AI in achieving environmental sustainability has been discussed in numerous studies in different sectors and contexts (Al-Sharafi et al., 2023; J. Chen & Wang, 2024). Examples are those related to sustainable farms (Bao & Xie, 2022; Murugan et al., 2022), sustainable food systems (Marvin et al., 2022), sustainable energy (Saheb et al., 2022), sustainable smart manufacturing (Tucker, 2021), sustainable marketing focused on green purchasing and decision aids (Frank, 2021), sustainable low-impact transportation (Budak & Sarvari, 2021), and the reduction of computational cycles and inference time in organizational information systems (Quan et al., 2024).

In line with the above, and despite their recent studies, the success of AI for sustainability still requires moving towards holistic metrics that combine technical and predictive performance models, measure the acceptance of target audiences, and are aware of the technical limitations of AI and the complexity of climate shocks (Nishant et al., 2020). But, as these authors state, to make meaningful progress towards environmental sustainability, it is necessary to have measurements and monitoring of interventions that are timely, accurate, and precise; it is also necessary for policy makers, organizations, and society to be able to clearly see their impact within organizational intervention strategies in relation to established goals, such as the nine PBs or the 17 SDGs.

Therefore, even though leveraging the advanced analytical capabilities of AI allows organizations to be more predisposed to identify areas for green innovation and design data-driven strategies to achieve these sustainability goals (Lin et al., 2024; Wang et al., 2023), the orientation of their managers towards it is key. Regarding this orientation, research has yet to deepen this discussion. Preliminary work, such as that of Mert et al. (2008), shows that for users the environmental/ecological reason is a positive side effect that affects the adoption of a new technology but does not outweigh the financial reason for using it. On the other hand, the study by Averdung and Wagenfuehrer (2011) argues that people who have a strong interest in protecting and improving the environment and consider sustainability to be important are more open to environmentally friendly technologies and consider them very useful. Likewise, work such as Pinski et al.'s (2024) argues that CEO orientation towards AI helps companies identify sustainable perspectives, foster a culture of sustainability, rationalize

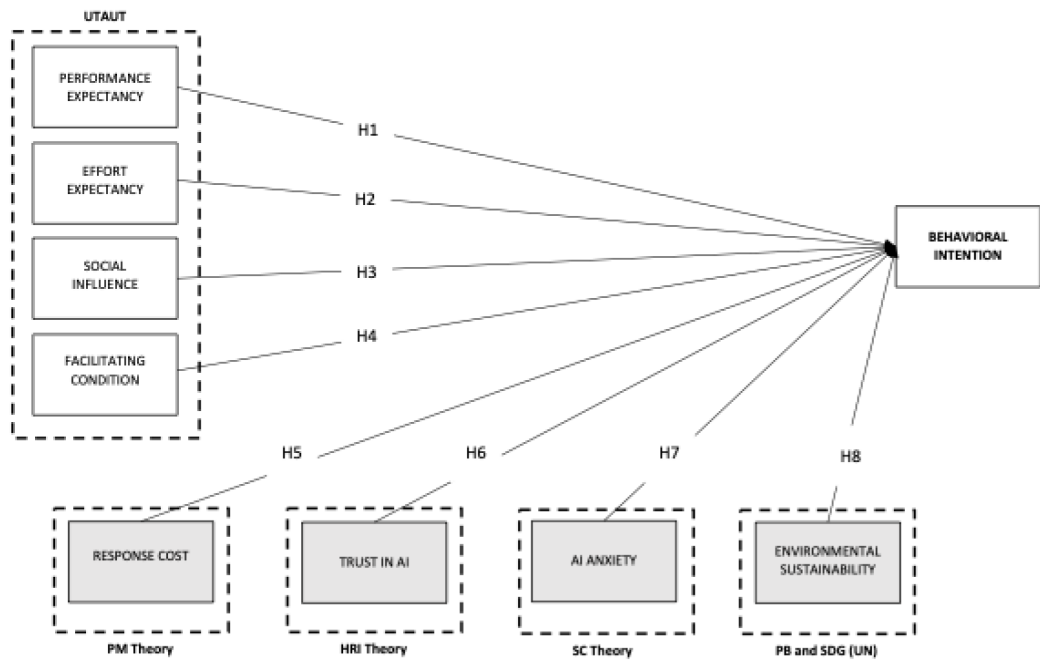
resource allocation, and enhance green innovation through judicious use of AI capabilities. Similarly, as recently stated by Wang and Wu (2024), for green innovation in Chinese manufacturing companies, CEO orientation towards AI will facilitate the adoption of these technologies for data analytics and predictive modeling, which will enable organizations to improve their understanding of environmental dynamics.

Based on the solid evidence establishing a relationship between these variables and the adoption of technological innovations, we propose the following hypothesis:

H8: Environmental sustainability is positively related to CEOs' intention to adopt AI tools in their companies.

As a result of the hypotheses explained above and to examine and understand the factors influencing this adoption, a new specific model is proposed, as seen in Figure 1.

Figure 1. AI acceptance model in companies for internal and productive processes



RESEARCH METHODOLOGY

To examine the factors influencing the adoption of AI in companies' production processes, an online survey was conducted with 409 CEOs and entrepreneurs from various organizations located in Spain. As reported by the National Observatory of Technology and Society (2023), although only 8% of Spanish companies have adopted AI, the country has reached an average level of AI adoption, comparable to the European Union average; this adoption level is still distant from the adoption levels of Denmark (24%), Portugal (17%), and Finland (16%). These figures highlight a significant margin for improvement and growth in AI adoption by Spanish companies in the coming years (National Observatory of Technology and Society, 2023). This technological context in which Spanish companies are immersed justifies the relevance of the study to clarify the determining factors for successful AI

implementation as well as to identify possible barriers and dilemmas associated with its integration into the productive and internal processes of these organizations.

Although the fieldwork was carried out during September and October 2023, the questionnaire was designed beforehand and was inspired by Venkatesh et al. (2003), and a pilot was conducted in two successive stages. The first involved reviewing the questionnaire with the support of other senior researchers to identify and resolve inaccuracies. Subsequently, 20 executives evaluated the revised version of the survey to ensure the quality and accuracy of the responses.

To collect the data, a self-administered form was distributed via email to the CEOs (in the case of large or medium-sized companies) or directly to the entrepreneurs (in the case of SMEs). The selected sample included CEOs and entrepreneurs, as Moore and Benbasat (1991) affirm that they are the final decision-makers when it comes to implementing technologies in their respective organizations.

The questionnaire used consisted of 39 questions and employed Likert scales that had been validated in previous research (Davis et al., 1989; McAfee & Brynjolfsson, 2012), ensuring the reliability and validity of the responses obtained in this study.

The CEOs' sample was obtained thanks to one of the researchers, who is also CEO of his own company and is attached to a national group of companies. Thanks to his group, both face-to-face and self-administered surveys were completed, and each CEO was in charge of surveying other CEOs in their sector, thus achieving the desired snowball effect. The result of the sample is shown in Table 1.

Table 1. Sample of CEOs

Size	Commerce	Industry	Technological	Services	Culture and Health	Total	%
from 0 to 9 workers	41	54	33	69	19	216	52.81%
from 10 to 49 workers	24	30	21	18	8	101	24.69%
from 50 to 249 workers	12	17	11	9	0	49	11.98%
> 250 workers	3	17	3	19	1	43	10.51%
Total	80	118	68	115	28	409	
%	19.56%	28.8%	16.63%	28.1%	6.85%		

RESULTS

For an adequate examination of the model through the partial least squares method, the reliability of the constructs and their respective measurement scales was first confirmed. This step is essential before evaluating the entire model. According to Henseler et al. (2015) and Roldán & Sánchez-Franco (2012), a factor loading of at least 0.7 was determined to be essential for constructs measured in mode A. These specific findings are detailed in Table 2.

Next, after confirming the factor loading, as shown in Table 3, the reliability of our constructs was examined using Cronbach's alpha and other composite reliability indicators. Following the suggestion of Hair et al. (2021), these indicators were established to exceed 0.7. Additionally, convergent validity was ensured by evaluating the average variance extracted, which exceeded the established criterion of 0.5 (Straub et al., 2004).

To confirm the discriminant validity of the latent variables, as shown in Table 4 and Table 5, both the Fornell and Larcker test and the heterotrait-monotrait approach, described by Henseler et al. (2015), were applied.

Furthermore, the importance of evaluating the R^2 of the higher-level constructs, as represented in Table 6, was highlighted. At the same time, as illustrated in Table 7, for a more comprehensive understanding of the structural model, the path coefficients and the explained variance of the endogenous variables were examined.

Table 2. Individual reliability of measurement scales (factor loadings)

	AI Anxiety	Behavioral Intention	Effort Expectancy	Environmental Sustainability	Facilitating Conditions	Performance Expectancy	Response Cost	Social Influence	Trust in AI
ANX1	0.823								
ANX2	0.866								
ANX3	0.904								
ANX4	0.749								
BI1		0.970							
BI2		0.979							
BI3		0.982							
BI4		0.979							
EE1			0.903						
EE2			0.946						
EE3			0.956						
EE4			0.924						
EE5			0.879						
ES1				0.942					
ES2				0.942					
ES3				0.925					
FC1					0.921				
FC2					0.935				
FC3					0.874				
FC4					0.865				
PE1						0.949			
PE2						0.943			
PE3						0.945			
PE4						0.963			
PE5						0.918			
PE6						0.936			
PE7						0.945			
RC1							0.854		
RC2							0.911		
RC3							0.814		
RC4							0.901		
SI1								0.857	
SI2								0.906	
SI3								0.883	

continued on following page

Table 2. Continued

	AI Anxiety	Behavioral Intention	Effort Expectancy	Environmental Sustainability	Facilitating Conditions	Performance Expectancy	Response Cost	Social Influence	Trust in AI
SI4								0.842	
SI5								0.854	
TS1									0.938
TS2									0.939
TS3									0.917

Note. ANX = AI Anxiety; BI = Behavioral Intention; EE = Effort Expectancy; ES = Environmental Sustainability; FC = Facilitating Conditions; PE = Performance Expectancy; RC = Response Costs; SI = Social Influence; TS = Trust.

Table 3. Composite reliability and convergent validity

	Cronbach's Alpha	Composite Reliability (rho_a)	Composite Reliability (rho_c)	Average Variance Extracted
ANX	0.856	0.864	0.903	0.701
BI	0.984	0.985	0.988	0.955
EE	0.956	0.957	0.966	0.850
ES	0.930	0.939	0.955	0.877
FC	0.921	0.922	0.944	0.809
PE	0.979	0.981	0.982	0.889
RC	0.822	0.853	0.892	0.735
SI	0.919	0.932	0.939	0.755
TS	0.924	0.925	0.952	0.868

Note. ANX = AI Anxiety; BI = Behavioral Intention; EE = Effort Expectancy; ES = Environmental Sustainability; FC = Facilitating Conditions; PE = Performance Expectancy; RC = Response Costs; SI = Social Influence; TS = Trust.

Table 4. Discriminant validity (Fornell-Larcker test)

	AI Anxiety	Behavioral Intention	Effort Expectancy	Environmental Sustainability	Facilitating Conditions	Performance Expectancy	Response Cost	Social Influence	Trust in AI
ANX	0.838								
BI	-0.258	0.977							
EE	-0.299	0.616	0.922						
ES	-0.142	0.439	0.607	0.936					
FC	-0.335	0.729	0.765	0.468	0.899				
PE	-0.286	0.626	0.752	0.710	0.614	0.943			
RC	0.368	-0.365	-0.111	0.086	-0.267	-0.078	0.857		
SI	-0.151	0.552	0.659	0.625	0.635	0.666	-0.055	0.869	
TS	-0.387	0.646	0.687	0.588	0.656	0.669	-0.172	0.632	0.931

Note. ANX = AI Anxiety; BI = Behavioral Intention; EE = Effort Expectancy; ES = Environmental Sustainability; FC = Facilitating Conditions; PE = Performance Expectancy; RC = Response Costs; SI = Social Influence; TS = Trust.

Table 5. Discriminant validity (heterotrait-monotrait ratio)

	AI Anxiety	Behavioral Intention	Effort Expectancy	Environmental Sustainability	Facilitating Conditions	Performance Expectancy	Response Cost	Social Influence	Trust in AI
ANX									
BI	0.278								
EE	0.327	0.635							
ES	0.169	0.456	0.643						
FC	0.372	0.765	0.815	0.504					
PE	0.312	0.635	0.779	0.743	0.644				
RC	0.432	0.396	0.126	0.119	0.298	0.099			
SI	0.168	0.571	0.701	0.684	0.686	0.703	0.080		
TS	0.438	0.677	0.730	0.633	0.711	0.703	0.199	0.685	

Note. ANX = AI Anxiety; BI = Behavioral Intention; EE = Effort Expectancy; ES = Environmental Sustainability; FC = Facilitating Conditions; PE = Performance Expectancy; RC = Response Costs; SI = Social Influence; TS = Trust.

Table 6. R² of the model

	R ²	R ² Adjusted
BI	0.653	0.646

Note. BI = Behavioral Intention.

Table 7. Structural model assessment (path coefficients)

	Original Sample (O)	P Values
H1. Performance Expectancy > Behavioral Intention	0.290***	0.000
H2. Effort Expectancy > Behavioral Intention	-0.078	0.324
H3. Social Influence > Behavioral Intention	0.001	0.980
H4. Facilitating Conditions > Behavioral Intention	0.448***	0.000
H5. Response Cost > Behavioral Intention	-0.233***	0.000
H6. Trust in AI > Behavioral Intention	0.238**	0.001
H7. AI Anxiety > Behavioral Intention	0.126**	0.001
H8. Environmental Sustainability > Behavioral Intention	-0.033	0.626

Note. ***p<0.001, **p<0.01, and *p<0.05 (based on one-tailed and bootstrap test with 10,000 samples).

Additionally, the importance of evaluating the R² of the higher-level constructs, as represented in Table 5, was also highlighted. At the same time, as illustrated in Table 6, for a more comprehensive understanding of the structural model, the path coefficients and the explained variance of the endogenous variables were examined.

Falk and Miller (1992) recommend that the explained variance (R²) for endogenous variables should exceed 0.1 to ensure adequate explanatory power. Our model demonstrates a robust R² value of 64.6%, significantly surpassing this threshold. This indicates that the proposed factors collectively explain a substantial proportion of the variance in CEOs' behavioral intention to adopt AI, highlighting the model's predictive strength. Additionally, considering the overall fit of the model, the standardized

root mean square residual indicator was used as a guide, which is especially useful for the partial least squares method. The model showed a value of 0.055, which is within the acceptable range according to Henseler et al. (2015).

Finally, the predictive capability of the model was evaluated, where the Q^2 values, presented in Table 8, reaffirmed its predictive relevance.

Table 8. Predictive relevance (Q^2 values)

	Q^2 Predict	RMSE
BI	0.629	0.611

Note. RMSE = Root Mean Square Error; BI = Behavioral Intention.

DISCUSSION AND CONCLUSIONS

Global information management represents a strategic pillar for modern companies, facilitating digital transformation and optimizing key organizational processes. In this context, the adoption of AI by top management is relevant to its management. This study identifies the determining factors from the CEOs' perspective, revealing the importance of integrating technological, emotional, and organizational approaches for effective adoption.

Our results highlight that the intention of CEOs to adopt AI in their companies' production processes is determined by facilitating conditions, performance expectancy, response costs, trust in AI, and AI anxiety. Specifically, this intention to adopt AI is explained by 64.6% (R^2) of the variance of all the data collected, exceeding the threshold suggested for considering that the model has explanatory relevance (Falk & Miller, 1992).

Of the explanatory variables in the UTAUT model, only Performance Expectancy and Facilitating Conditions explain the adoption of AI by CEOs. Corroborating recent research by Alonso (2023) and Jain et al. (2022), which highlight performance improvements in internal processes as catalysts for AI adoption in companies, our findings reveal the significant impact of performance expectancy on the intention to adopt AI. Similarly, facilitating conditions also demonstrated a positive and significant effect, underscoring the importance of having adequate infrastructure and support for AI adoption, as highlighted in the literature (Behl et al., 2022; Sood et al., 2022). In contrast to what was demonstrated by Wang and Su (2021), Bodea et al. (2020), and Habila et al. (2023), factors such as effort expectancy and social influence did not show statistical significance as factors justifying the adoption of AI in the intention to use AI in corporate internal processes.

As for the four specific factors that have been used for further adaptation to the object of study, three of them have proved significant in improving the understanding of this adoption. The first of them, response costs, aligned with the research by various authors (Milne et al., 2000; Rainear & Christensen, 2017; Quan et al., 2024; Ye et al., 2024) and demonstrates that financial challenges, required time, and difficulties associated with AI implementation are factors that explain the intention to use AI in organizations. The fact that response costs emerged as one of the most significant factors in our study confirms the necessity for companies to carefully consider the costs associated with AI adoption (Chen, L. et al., 2023). As is the case for other technology adoptions, such as blockchain (Marikyan et al., 2022) or payments using mobile apps (Hamzah, 2024) or ChatGPT (Hung & Chen, 2023), in our study for CEOs the response costs of AI have a negative impact on usage intention. However, the fact that they perceive that AI can improve global information management—understood as organizational knowledge management—customer relations, or process optimization, opens up the possibility that they can overcome these barriers, both economic (response costs) and emotional (anxiety).

In line with the above, the second factor, AI anxiety, was found to hinder the intention to use this technology. Agreeing with Budhwar et al. (2022), although companies today seem to be getting accustomed to this new technology, it does not mean they are free from experiencing some anxiety about its impact on their internal processes. The statistical significance of this factor aligns with studies that identify it as a barrier to technological adoption (Budhathoki et al., 2024; Gaudiello et al., 2016; Thatcher & Perrewé, 2002). In line with our results, Suseno et al. (2022) also warn about the anxiety that managers have when preparing the shift towards the adoption of AI within organizations. Also, Sajdak and Młody (2024) state that technological anxiety hinders the development of a company and the maintenance of a competitive advantage over time by generating aspects such as resistance to change, lack of management support, and incompatibility in the value chain or fear of failure.

The third significant factor, trust in AI, also proved to be significant, emphasizing the importance for CEOs to understand and trust the capabilities of this technology for its effective implementation in their companies' processes. It is worth noting that in our study, CEOs perceived anxiety as well as trust as factors that influenced their adoption. The results are consistent with what the literature shows about how the distrust top managers have in technologies for organizational processes comes from anxiety about how to manage it internally (Hengstler et al., 2016). In line with this, previous studies such as Sidlauskiene's (2022) and McKnight et al.'s (2011) support trust as an influential factor in adoption that contributes to changes in usage intention (Chi et al., 2021; Tussyadiah et al., 2020). Our results extend Majrashi's (2024) recent evidence by finding that perceived trust on the part of an organization's decision makers goes beyond the public sector by influencing the perception of private companies' CEOs for their internal processes.

Given the above, factors such as confidence in AI and technology anxiety shape CEOs' willingness to adopt AI in production processes. These factors are key not only to establishing priorities in global information management, but also to fostering an organizational culture that values the strategic use of AI in critical areas such as process optimization and data-driven decision-making; in this sense, CEOs' leadership and attention to innovation and their orientation towards AI are crucial to ensuring the implementation and sustainability of these technologies in companies (Ye et al., 2024).

Finally, although Pinski et al. (2024) and Wang and Wu (2024) affirm the importance of AI-oriented managers to facilitate green innovation and environmental sustainability in their internal processes, our results do not show that CEOs, in the Spanish context, consider this factor as relevant when adopting this technology to design strategies that help them achieve their sustainability goals. The perception of these CEOs detracts from what authors such as Lin et al. (2024) and Wang et al. (2023) state when they affirm that AI contributes to designing these sustainable strategies. An example of this is the role of AI in optimizing information generated in supply chains and environmental processes (Huang et al., 2024; Tsang et al., 2024). However, despite the role of AI in global information management, with the evidence from our study and that of Mert et al. (2008), for CEOs the financial or cost rationale is more important than the environmental sustainability rationale. Therefore, unfortunately, in the case of CEOs of Spanish companies, these environmental factors are not a critical factor in their decision-making when incorporating it to improve the management of global information in different areas of their companies, ranging from problem-solving (Almuraqab et al., 2024) to predictive scenario analysis or optimization of the logistics and production processes (Huang et al., 2024; Na et al., 2023).

In summary, the results of this study highlight that the adoption of this technology is influenced by the aforementioned factors, while traditional factors such as Effort Expectancy and Social Influence are no longer relevant to its adoption. Consistent with Mariani et al. (2023), these findings reinforce the understanding that AI adoption in companies requires a more pragmatic and contextualized analysis, as is the case in the context of its adoption in the internal processes of organizations. Therefore, integrating AI into global information management requires a strategic approach that balances costs, anxiety, and perceived benefits for CEOs. By connecting these findings with existing literature, this study reinforces the need for specific models that address the particularities of business leaders as

key players in digital transformation. This not only drives the adoption of advanced technologies, but also establishes a framework for more informed and sustainable decision-making in a globalized environment by avoiding biases and helping to mitigate suboptimal decisions stemming from CEOs' subjectivity, fears, or hindsight bias (Hasan et al., 2022).

Theoretical, Practical, and Policy Implications

From a theoretical perspective, this study proposes a solid framework for analyzing, for the first time, the drivers that motivate CEOs to adopt AI in their internal processes. Considering organizational, technological, and human variables, this model demonstrates its capacity to capture the particularities of this technological adoption phenomenon. Specifically, it encourages more innovative scientific debate on how response costs and anxiety enrich the explanatory capacity of classic models of technology adoption in the context of organizational processes and from the perspective of CEOs.

On the other side, although different factors of AI trust have been addressed (Alsheibani et al., 2019; Majrashi, 2022; McKnight et al., 2002), this is the first work that addresses the importance of CEOs' trust as a significant factor for AI's adoption in firms.

Regarding the practical implications, this study provides the business fabric with knowledge about which aspects their CEOs should evaluate beforehand before implementing AI in the internal processes of their companies, specifically what competitive advantages AI will offer their company, how it will increase productivity, and how it will promote stronger leadership in their respective sectors. However, despite the many advantages, according to some aspects revealed by this work, it is recommended that CEOs be cautious and provide the necessary resources, training, and information before incorporating AI into their company's processes to reduce fear and anxiety towards AI. Reducing this anxiety will help generate sufficient trust for better technological integration and smoother internal use. Finally, making companies aware of the importance of early detection of response costs associated with AI adoption will allow them to mitigate its impact. In this sense, it will help them anticipate and implement proactive strategies to minimize these costs before implementation.

For these reasons related to policy implications, the results obtained show that the CEOs are willing to implement it as long as the costs of its development and maintenance in their internal processes are not too high. This highlights the importance of incorporating public policies for the technological transformation of the different industries in a country, developing a subsidy plan that reduces this barrier and promotes a business awareness that these technologies facilitate the environmental sustainability so much in demand by society as a whole.

Limitations and Future Research Directions

Although this study provides a novel perspective on the intention to adopt AI among senior company executives, it leaves open the question of how AI can be adopted. Additionally, to better understand how AI can be adopted, it is recommended that future research incorporate qualitative analyses to enrich the empirical study of AI adoption in various business areas.

Moreover, while this study offers important insights into the Spanish business context, it has a limitation. Therefore, it would be interesting to apply the research to other regions and markets to evaluate the generalizability of the results obtained. Conducting cross-cultural studies would allow for a deeper understanding of the particularities in AI adoption in business environments across different continents.

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APPENDIX

Table 9. Measurement scales

Construct	Scale	Sources
Performance Expectancy	PE1: I believe AI tools are useful for optimizing our company's internal and productive processes. PE2: I believe AI tools can execute our company's internal and productive processes more efficiently. PE3: I believe AI tools can increase productivity in our company's internal processes. PE4: I believe AI tools would improve performance in our company's productive processes. PE5: I believe AI tools can provide more detailed information about our internal processes. PE6: I believe AI tools will enhance the quality of information related to our company's productive processes. PE7: I believe AI tools will provide new valuable information that can improve our internal and productive processes.	(Mcafee & Brynjolfsson, 2012; Moore & Benbasat, 1991; Venkatesh et al., 2003)
Effort Expectancy	EE1: Using AI tools in internal and productive processes would be clear and understandable to our company's staff. EE2: It would be easy for our company to familiarize itself with AI tools for optimizing internal and productive processes. EE3: It would be easy for our company to use AI tools in the optimization of internal and productive processes. EE4: I believe learning about AI tools would be easy for our company's staff when implementing them in internal and productive processes. EE5: We could generate valuable data using AI tools in internal and productive processes in a simple manner.	(Venkatesh et al., 2003)
Social Influence	SI1: Companies that influence ours use AI tools to optimize internal and productive processes. SI2: Companies we consider as references use AI tools in their internal and productive processes. SI3: Companies in our environment that use AI tools to optimize processes have more prestige than those that do not. SI4: Companies in our environment that use AI tools in their processes are considered innovative. SI5: Using AI tools for process optimization is a status symbol in our environment.	(Venkatesh et al., 2003)
Facilitating Conditions	FC1: Our company has the necessary resources to use AI tools in the optimization of internal and productive processes. FC2: Our company has the necessary knowledge to use AI tools in improving internal and productive processes. FC3: AI tools for process optimization are compatible with other systems in our company. FC4: Our company has a person (or group of people) to provide assistance if difficulties arise with AI in process optimization.	(Venkatesh et al., 2003)
Response Costs	RC1: Implementing AI tools in internal and productive processes is costly in terms of purchase and operation. RC2: We need to frequently update our AI tools implemented in internal and productive processes. RC3: If there are security issues in the company, they can affect the efficiency of the AI tools implemented in our company. RC4: Complying with the security policy of AI tools in our company would require considerable effort beyond time.	(Al-Emran et al., 2020; Johnston & Warkentin, 2010)

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Table 9. Continued

Construct	Scale	Sources
Trust in AI	TS1: I generally trust AI tools until they prove themselves to be untrustworthy. TS2: I usually give AI tools the benefit of the doubt when implementing them in our company for the first time. TS3: I usually trust an AI tool until it gives me a reason to distrust it.	(Mcknight et al., 2011)
AI Anxiety	ANX1: I am concerned about implementing AI tools in our company's internal processes. ANX2: I am afraid that I might lose company information by incorrectly using AI tools in internal processes. ANX3: I hesitate to use AI tools in productive processes for fear of making mistakes that the company cannot correct. ANX4: Using AI tools in the company's internal processes feels challenging to me.	(Rana et al., 2017)
Environmental Sustainability	ES1: Using AI tools can help us save various resources in our productive processes (e.g., energy and water). ES2: Using AI tools can help us save on production costs. ES3: Using AI tools can help us better manage waste in our internal or productive processes.	(Baudier et al., 2020; Gansser & Reich, 2021)
Behavioral Intention	BI1: We intend to use AI tools to attract customers in the coming months. BI2: We predict that we will use AI tools in customer acquisition in the coming months. BI3: We plan to use AI tools to attract customers in the coming months. BI4: We intend to obtain new and valuable data in customer acquisition thanks to AI tools in the coming months.	(Davis et al., 1989; Venkatesh et al., 2003)

Note. PE = Performance Expectancy; EE = Effort Expectancy; SI = Social Influence; FC = Facilitating Conditions; RC = Response Costs; TS = Trust; ANX = AI Anxiety; ES = Environmental Sustainability; BI = Behavioral Intention.

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